

2b. Portfolio Theory

- estimation risk - historical estimated parameters differ from historical true parameters
 - realized returns are very noisy proxies for $E(r)$
 - with many stocks, covariance errors happen often (100 stocks = 4950 covariances)
 - to help reduce this, estimate using portfolios rather than individual stocks
- Model risk - test for structural breaks in the data
- Merton formula - when we can invest in multiple risky assets and borrow at the risk free rate:
 - invest if: $\text{Sharpe ratio}_i > \text{Sharpe ratio}_p$
 - and disinvest if the opposite is true

2c. Capital Asset Pricing Model

9/18/25

9/23/25

- equilibrium when supply of security X = market demand for X
- goal for portfolio of risky assets: have the highest Sharpe ratio
 - adjust Sharpe by adding/losing small amts. of optimal securities

- CAPM is an equilibrium model: what investors want to hold must equal what they do hold (the market portfolio)
- market portfolio has highest possible Sharpe ratio
 - $E(r_i) = r_f + \beta_i(E(r_m) - r_f)$, where $\beta_{im} = \text{Cov}(r_i, r_m) / \sigma_m^2$
 - essentially: if this (the CAPM model) holds, we cannot improve the market.
 - beta β is the relevant measure of risk for CAPM. higher = riskier

- Covariance math for CAPM:

$$\rho_{ij} = \sigma_{ij} / \sigma_i \sigma_j \quad \text{order doesn't matter}$$
$$\beta_{ij} = \sigma_{ij} / \sigma_j^2 = \rho_{ij} \left(\frac{\sigma_i}{\sigma_j} \right) \quad \text{order does matter}$$

- in CAPM assumptions, volatility doesn't matter. it dissipates because it is idiosyncratic risk. $E(r)$ directly equals/correlates to β , risk.